CST-452 Senior Capstone | Final Project Presentation

EERoom

Electrical engineering simulations that run in a browser



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www.eeroom.io

The problem: learning engineering only on paper

1

Concepts stay abstract

Ohm's Law, sampling, and Boolean logic are usually taught as equations. Students learn faster when they can change a value and see the result.

2

Tools are hard to reach

Industry software such as MATLAB, Ansys, and Cadence is licensed per seat and installed on lab computers.

3

Steep learning curve

Professional tools take hours to learn before a student can try a simple idea.

Goal: free, focused tools that open in any browser, need no install, and let students save and share their work.

The solution: EERoom

1

Ohm's Law and Power Calculator

Enter any two of V, I, and R. The third value and the power update as you type.

2

Waveform Visualizer

Four signal shapes with a sample rate control that shows aliasing as it happens.

3

Logic Gate Sandbox

Seven gates on a grid with a live truth table and a Boolean expression for each output.

Project portfolio: www.eeroom.io/portfolio

Platform features

- Use every tool as a guest, no account needed
- Free account to save, reopen, and delete projects
- Password reset by email
- Read-only share links anyone can open
- Works on desktop and tablet
- Live at www.eeroom.io

Logic Gate Sandbox



- AND, OR, NOT, NAND, NOR, XOR, and XNOR on a click-to-place grid
- Wire by clicking or dragging between pins
- Truth table and Boolean expression update as you build
- Unconnected pins are flagged, not hidden
- Feedback loops are refused instead of guessed

Shown: a half adder, Y = A XOR B and Z = A AND B

Waveform Visualizer and aliasing



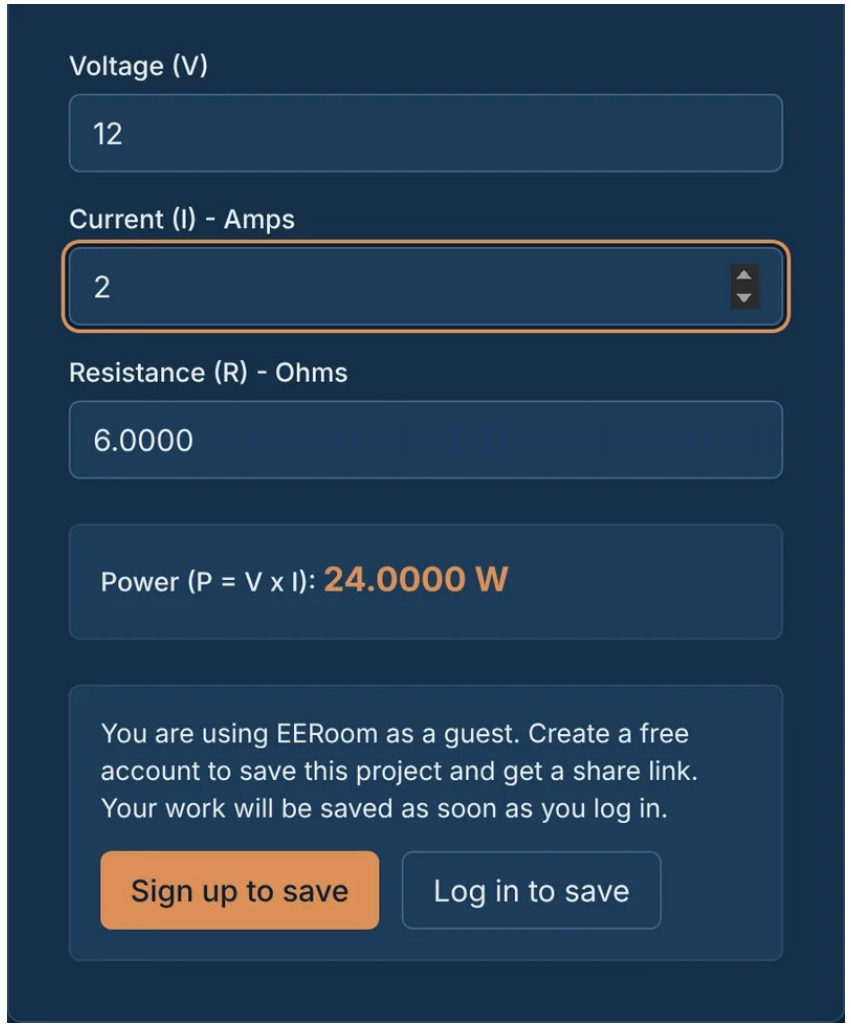
Shown: a 50 Hz sine sampled at 80 Hz. The red trace is the alias the samples suggest.

$$f_s \leq 2f$$

When the sample rate is at or below twice the signal frequency, the samples describe a false, slower wave.

- Sine, square, sawtooth, triangle
- Amplitude, frequency, phase, DC offset
- RMS, peak-to-peak, and period readouts
- Ideal and sampled traces overlaid

Ohm's Law calculator and guest access



Calculator

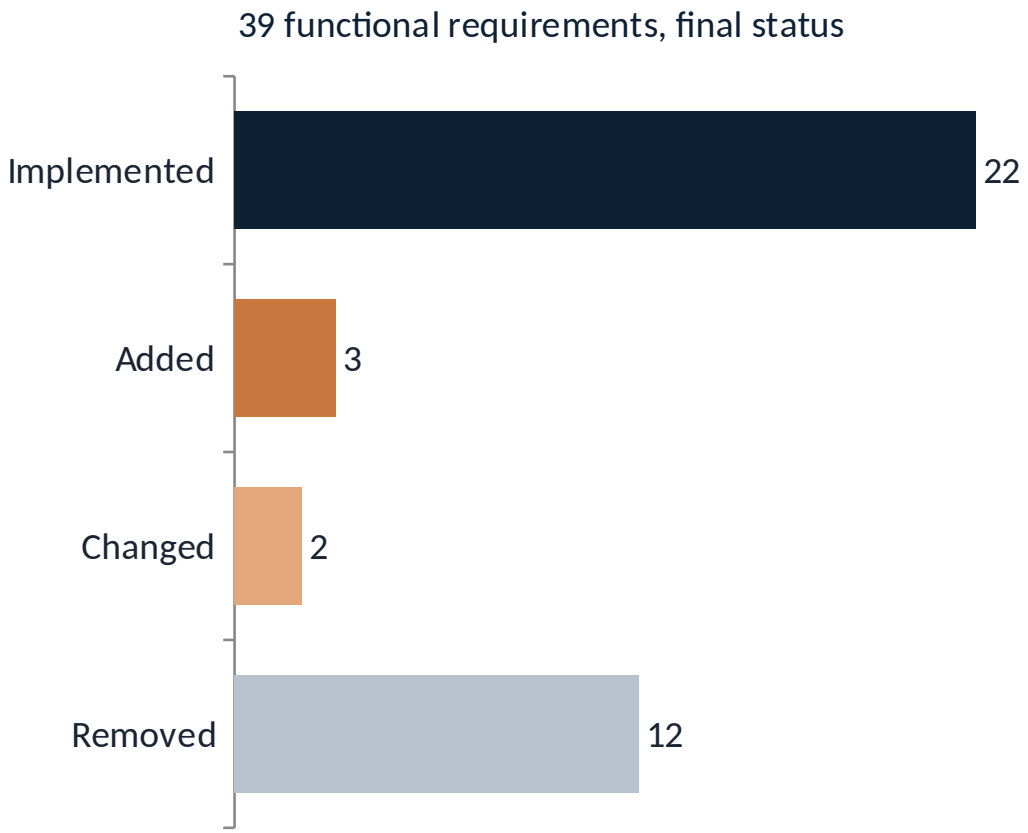
Enter any two values and the third and the power follow. 12 V and 2 A give 6 Ω and 24 W. Zero values that would divide by zero show a clear message.

Guest access, added in iteration 3

Use any tool as a guest → Click Sign up to save → Register and log in → Work is saved and reopened

The work is held in session storage during sign-up and saved exactly once.

Requirements and scope decisions



Removed after the first iteration

DC circuit simulator

Highest risk; could not be finished and tested in time

Google sign-in

Email login already met the requirement

Stripe Pro tier

Payments added security risk, not learning value

Every change was recorded with its reason and submitted for approval, and the time saved went into testing, security, and accessibility.

Architecture

React client

Vercel, www.eeroom.io

- React 18, React Router, Tailwind
- Logic and waveform engines
- SVG board, canvas display

Express REST API

Railway

- /auth (rate limited)
- /projects (JWT required)
- /share/:token (public)

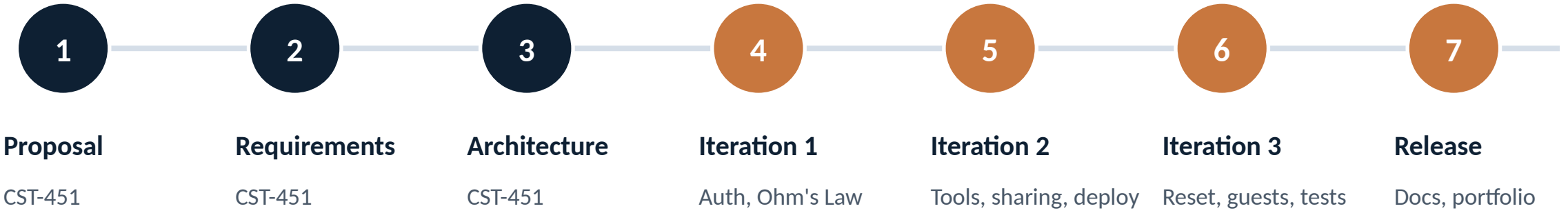
Supabase

PostgreSQL and Auth

- users and projects tables
- bcrypt password hashing
- Password recovery email

- The client calls the API over HTTPS with a signed JWT; the API talks to Supabase
- All simulation math runs in the browser, so results are instant
- Each tool saves its state as JSON in one projects table, so a new tool needs no database change
- Every push to GitHub deploys the client and the API automatically

Design process: seven stages



● Planning (CST-451) ● Build, test, release (CST-451 and CST-452)

Kanban backlog

20 items with estimates and actual hours

Traceability

Every requirement mapped to code and a test

Scope review

Changes recorded and justified each iteration

Security built into the design

Passwords

Hashed with bcrypt by Supabase Auth; never stored by EERoom

Sessions

Signed 24-hour JWT checked on every project request

Ownership

Every query filtered by the user ID from the token

Brute force

10 login attempts per 15 minutes per address

Account privacy

Login and reset never reveal whether an email exists

Sharing

64-character random tokens; no private fields returned

Testing: 119 automated tests, all passing

82

frontend unit tests

Logic evaluator, waveform math, reset links, guest drafts

37

API integration tests

Real HTTP requests through Express with Supertest

27

manual test cases

Run against the live site, all passed

Proving the tests catch real bugs

I deliberately removed the ownership check from the project API. The test that opens another user's project failed right away, which confirmed the suite detects a real security defect. Then I put the check back.

Challenges and lessons learned

Scope was too big

Cutting scope early, on purpose, produced a finished and tested product instead of a half-working larger one.

Deployment surprises

CORS, environment variables, the proxy behind the rate limiter, and email redirects only showed problems on real hosts. Deploy early.

Testing needs design

Keeping the math in pure functions and splitting the Express app from the server start made both easy to test.

Security is in the details

Same answers for unknown emails, ownership filters, and rate limits are small pieces of code that close real holes.

Purpose and next steps

Christian worldview

EERoom is built as an act of stewardship and service: free tools that give every student the same chance to practice, regardless of cost or school. Colossians 3:23 shaped the work itself. Doing it well meant finishing fewer features properly, protecting users' data, and being honest about limits.

Future work

- DC circuit simulator using nodal analysis
- Phone layouts below 768 px
- More logic components, such as flip-flops
- Rooms for other disciplines, such as mechanical and software engineering
- Classroom features so instructors can share assignments

“Whatever you do, work heartily, as for the Lord and not for men.” Colossians 3:23 (ESV)

Live demonstration

Thank you

Live application

www.eeroom.io

Project portfolio

www.eeroom.io/portfolio

Source code

github.com/AidenVargas03/EERoom



Scan to try it